Programming Homework: Support Vector Machines

Problem 1: Comparison of Linear and Non-linear SVM

Requirements:

Implement a program comparing the performance of different SVM types on synthetic and real datasets, using the `scikit-learn` library.

a) Data generation and preprocessing:

- Create a 2D synthetic dataset using `make_classification` (500 samples, 2 classes)
- Prepare a real-world dataset: Iris dataset (binary classification)
- Split data into train/test sets (70/30 ratio)
- Normalize data using `StandardScaler`

b) Train and evaluate SVM models:

- Linear SVM (`LinearSVC`)
- SVM with RBF kernel (`SVC(kernel='rbf')`)
- SVM with polynomial kernel (`SVC(kernel='poly', degree=3)`)
- SVM with sigmoid kernel (`SVC(kernel='sigmoid')`)

c) Analyze results:

- Compare accuracy on the test set

- Count the number of support vectors for each model
- Plot the decision boundary for each method
- Compare training time

d) Explain performance differences among kernels and identify scenario-based kernel selection guidelines.

Hints:

- Use `sklearn.svm` for SVM models
- Use `sklearn.metrics` for performance evaluation
- Use `matplotlib` to visualize decision boundaries
- Use the `time` module to measure training time

Problem 2: Study on the Effect of SVM Parameters

Requirements:

Investigate the influence of key SVM parameters and soft-margin SVM techniques.

a) Study the effect of the parameter C (regularization parameter):

- Experiment with C values: [0.001, 0.01, 0.1, 1, 10, 100, 1000]
- Plot graphs illustrating the relationship between C and:
  - Accuracy on train and test sets
  - Number of support vectors
  - Margin width
- Analyze overfitting/underfitting phenomena

b) Study the effect of the gamma parameter in RBF kernel:

- Experiment with gamma values: [0.001, 0.01, 0.1, 1, 10, 100]

- Analyze the effect on decision boundary complexity
- Determine the optimal gamma value for the problem

c) Compare hard-margin and soft-margin SVM:

- Generate noisy data using `make_blobs` with `cluster_std=1.5`
- Compare performance of SVM with C=1 (soft) and C=1000 (hard)
- Analyze the number of margin-violating points and support vectors

d) Hyperparameter optimization using GridSearchCV:

- Search for the best combination of C and gamma
- Compare results before and after optimization
- Evaluate performance using cross-validation

Hints:

- Use `GridSearchCV` or `RandomizedSearchCV` for parameter optimization
- Use `learning_curve` to analyze the bias-variance tradeoff
- Use `validation_curve` to study the effect of individual parameters
- Visualize results using `matplotlib`